

## Setup the required tools

The provided commands builds the following required tools:

1. **gdb-gef** (enhanced version of gdb)
  - uses the same basic commands as the **gdb** tool, with additional ones (refer to cheatsheet)
2. **pwntools** (Python library)
  - eg. **pwn cyclic ...**

Ensure that you are running these commands from the same directory as the **Dockerfile** file

```
docker build -t pwn-tools .
```

```
docker run -it --cap-add=SYS_PTRACE --security-opt seccomp=unconfined pwn-tools
```

You will be provided with a terminal environment that looks similar to this:

- the **ret2sentctf** binary will be available
- you may directly run **gdb** on it

```
root@4586449e4d10:/workspace#
```

```
root@4586449e4d10:/workspace# ls
ret2sentctf
root@4586449e4d10:/workspace# gdb ret2sentctf
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
GEF for linux ready, type `gef' to start, `gef config' to configure
90 commands loaded and 5 functions added for GDB 12.1 in 0.00ms using Python engine 3.10
Reading symbols from ret2sentctf...
(No debugging symbols found in ret2sentctf)
gef>
```